

Table 1: Direct comparison of original and PT2-expanded Siamese+MLP Raman+FT results on the original real-mixture dataset

Setup	Class space	Evaluation protocol	Precision	Recall	Micro-F1	Avg. pred. chemicals
Original <code>Siamese_MLP_6_FFT_Threshold</code> calibrated	12-class head	25% calibration / 75% held-out test, calibrated thresholds	0.970	0.980	0.970	2.02 ^a
PT2-expanded retrain, default scoring	17-class head	Full original real set, all 17 classes counted, fixed threshold 0.5	0.597	0.888	0.714	2.97
PT2-expanded retrain, notebook-style filtered	17-class head	Full original real set, report filtered to original supported classes only, fixed threshold 0.5	0.872	0.888	0.880	2.97
PT2-expanded retrain, recalibrated	17-class head	25% calibration / 75% held-out test on original real set, notebook-style filtered report, recalibrated 17-class thresholds	0.774	0.936	0.847	3.36

The original calibrated row is taken from `Siamese_MLP_6_FFT_Threshold.ipynb`.

^a The original notebook did not log mean predicted-label count explicitly; 2.02 is inferred from micro precision and recall, with mean true labels fixed at 2.0.

“Notebook-style filtered” means the report only includes classes with nonzero support in the original real-mixture set, matching the notebook logic and ignoring false positives on absent classes.

The recalibrated PT2-expanded row uses newly fit 17-class thresholds from the new directory `pt2_augmented_existing_real_recalibrated`; its exact filtered micro metrics are precision 0.7738, recall 0.9356, micro-F1 0.8470.

Table 2: Per-chemical precision, recall, and F1 on the eight original supported chemicals

Chemical	Original calibrated			PT2-expanded filtered 0.5			PT2-expanded recalibrated		
	P	R	F1	P	R	F1	P	R	F1
1-dodecanethiol	0.87	0.99	0.93	0.858	0.893	0.875	0.860	0.978	0.915
6-mercapto-1-hexanol	0.94	0.93	0.93	0.330	0.333	0.332	0.271	0.951	0.422
benzene	1.00	1.00	1.00	1.000	1.000	1.000	1.000	0.967	0.983
benzenethiol	1.00	0.98	0.99	1.000	1.000	1.000	1.000	0.943	0.971
etoh	1.00	0.98	0.99	0.741	0.992	0.848	0.974	0.804	0.881
meoh	1.00	0.98	0.99	1.000	0.872	0.932	1.000	0.879	0.936
n,n-dimethylformamide	1.00	1.00	1.00	1.000	1.000	1.000	1.000	1.000	1.000
pyridine	1.00	1.00	1.00	1.000	1.000	1.000	1.000	1.000	1.000

The original calibrated per-chemical metrics are from the held-out 75% test report in `Siamese_MLP_6_FFT_Threshold.ipynb`.

The PT2-expanded filtered 0.5 row uses the full original real set but only reports the eight classes with nonzero support in the original dataset, matching the notebook’s reporting convention.

The PT2-expanded recalibrated row uses the new 17-class thresholds fit on the 25% calibration split; the main gain is recall, but it comes with a large precision collapse for 6-mercapto-1-hexanol.